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SIMATS Online Programming
Lab

JAVA PROGRAMMING

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CO5-AT3-Level Exam3

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Level Exam

Java

03 Sep 2026, 12:01 PM

QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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Level 3 — Hard (H) Permutations — Numbers

✓ Submitted

Write a program to print unique permutations of a given number

Sample Input:

Given Number: 143

Sample Output:

Permutations are:

134

143

314

341

413

431

Testcase: 1. 0

2. 111

3. 505

4. -143

5. -598

Your Code

```
1 import java.util.*;  
2 class Main{
```

```
3 public static void main(String[] args){
4     Scanner sc = new Scanner(System.in);
5     String n = sc.next();
6     permute(n, " ");
7 }
8 static void permute(String s,String ans){
9     if(s.length()==0){
10         System.out.println(ans);
11         return;
12     }
13     for(int i=0;i<s.length();i++){
14         char ch = s.charAt(i);
15         if(s.indexOf(ch)!=i)
16             continue;
17         String rem = s.substring(0,i)+s.substring(i+1);
18         permute(rem,ans+ch);
19     }
20 }
21 }
```

Output

```
143
134
413
431
314
```

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QUESTIONS

LEVEL 3 — HARD

Q1
Numbers**Q2**
Control

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Level 3 — Hard (H) Steps — Control **Submitted**

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase:1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

Your Code

```
1 import java.util.*;
```

```
2 class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n=sc.nextInt();
6         int count=0;
7         while(n>0){
8             if(n%2==0)
9                 n=n/2;
10            else
11                n=n-1;
12            count++;
13        }
14        System.out.println(count);
15    }
16 }
```

Output

6



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